

Unlocking Latent Clinical Knowledge with Parameter-Efficient Fine-Tuning

A QLoRA Approach to In-Hospital Mortality Prediction on Constrained Hardware

Thomas I. Bales

Abstract

Early identification of ICU patients at high risk of in-hospital mortality supports triage and timely intervention, but deploying predictive models in care settings demands more than raw accuracy: it demands reproducibility, auditability, and an honest account of cost. We ask whether a small, open foundation model—*llama-3.1-8B*—can be made to *beat its own untuned self* on a specialized clinical task using a parameter-efficient adapter of only 6.8M trainable parameters (0.08% of the model), trained end to end on a single 16 GB GPU. Using patient descriptions derived from MIMIC-III and 4-bit QLoRA fine-tuning, and after finding and removing a critical validation-set leak, the adapted model improves held-out recall from 0.18 to 0.46, precision from 0.11 to 0.56, and accuracy from 74.5% to 90.0% on 200 patients never seen during training—gains on both axes at once. We characterize the recall–precision oscillation observed during training as a clinical-*values* decision rather than a technical default, trace the performance ceiling to conflicting labels in coarse features, and argue that deterministic (temperature 0.0) inference is a prerequisite for clinical auditability. All results were produced at near-zero marginal energy cost, foregrounding a governance question about where the energy of this capability was actually spent.

```
<ccs2012> <concept><concept_id>10010147.10010257</concept_id>  
<concept_desc>Computing methodologies Machine learning</concept_desc>  
<concept_significance>500</concept_significance></concept>  
<concept><concept_id>10010405.10010481</concept_id> <concept_desc>Applied  
computing Health informatics</concept_desc>  
<concept_significance>500</concept_significance></concept> </ccs2012>
```

Introduction

In-hospital mortality is one of the oldest and most consequential prediction targets in critical care. A model that flags, early in an admission, which patients are at elevated risk of dying can inform triage, prompt escalation of care, and focus scarce clinician attention where it matters most. Because a missed death is far costlier than a false alarm, the metric that matters is not overall accuracy but *recall*—and recall is precisely what a naive classifier sacrifices, since at an 11% base rate a model that predicts “survives” for everyone already scores 89% accuracy while catching nobody.

Two forces make this an interesting problem in 2026 rather than a solved one. First, open foundation models such as *llama-3.1-8B* now encode a great deal of latent clinical knowledge absorbed during pretraining. Second, parameter-efficient fine-tuning (PEFT) makes it possible to adapt such a model with a tiny fraction of its parameters, on

hardware a research group or a hospital IT department already owns. The natural question—and the “high-risk” bet of this project—is whether a small, general, open model can be made to *outperform its own zero-shot baseline* on a specialized clinical task, using an adapter that trains in a few hours on a single GPU, and whether the result can be made reproducible and auditable enough to be defensible in a care setting.

This framing is deliberately about more than a leaderboard number. Deploying a predictive model in healthcare raises three questions that a benchmark hides: (i) *at what operating point* should the model run, given that catching more deaths always means more false alarms; (ii) *what is the model actually learning*, and where does the data stop it; and (iii) *what did the capability cost*, and who paid for it. We treat these not as afterthoughts but as first-class results.

Contributions.

This paper reports: (1) an end-to-end QLoRA fine-tuning pipeline for mortality prediction that runs on a single 16 GB GPU; (2) a rigorous held-out evaluation made trustworthy only after discovering and removing a validation-set leak that would have invalidated the headline claim; (3) a characterization of the recall–precision operating frontier as a clinical-values decision; (4) identification of a data-resolution ceiling caused by conflicting labels in coarse features; and (5) an honest account of the memory and thermal constraints that dominated the engineering effort.

Related Work

Parameter-efficient fine-tuning.

LoRA freezes a pretrained model and injects trainable low-rank matrices into its attention projections, cutting the number of trainable parameters by orders of magnitude while matching full fine-tuning quality on many tasks. QLoRA extends this by quantizing the frozen base model to 4 bits (NF4) and back-propagating through it, enabling fine-tuning of models that would not otherwise fit in a single GPU’s memory. Both works address the same obstacle this project faced—adapting a large model under a hard memory budget—and our method is a direct application of their techniques to a clinical setting.

Language models for clinical prediction.

Rasmy et al.’s Med-BERT pretrains a transformer on structured diagnosis codes from large EHR corpora and shows gains on downstream disease prediction, establishing that clinical-domain pretraining transfers to prediction tasks. Jiang et al. scale this idea to a health-system deployment (NYUTron), fine-tuning a language model on unstructured clinical notes to predict outcomes including in-hospital mortality and readmission, and report performance competitive with structured models. Both rely on large-scale clinical pretraining or health-system-scale infrastructure. Our work asks a complementary question: how far can a *small, general, open* model plus a 0.08% adapter get on

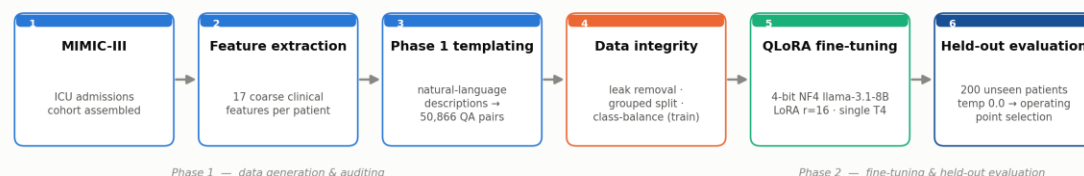
commodity hardware, and it foregrounds the governance dimensions those deployment-focused papers necessarily treat as secondary.

Data foundation.

All patient records derive from MIMIC-III, a de-identified critical-care database that has become a standard benchmark substrate for clinical machine learning. We use a coarse, 17-feature view of each admission (see Section 3), which—as we show—is both the source of the model’s efficiency and the source of its ceiling.

Methodology

Figure 1 summarizes the end-to-end workflow, from raw MIMIC-III admissions through data auditing to held-out evaluation. We describe each stage below.



End-to-end workflow. Phase 1 assembles a cohort from MIMIC-III, reduces each admission to 17 coarse features, and templates it into a natural-language description with a one-token Yes/No mortality answer (50,866 pairs). Phase 2 audits the data for leakage and label conflicts, fine-tunes a 4-bit Llama-3.1-8B with a LoRA adapter on a single T4 GPU, and evaluates on 200 held-out patients at temperature 0.0. The orange stage (data integrity) is where the most consequential correction was made.

Data and cohort

Each ICU admission in the cohort is reduced to 17 coarse clinical features (e.g., age band, gender, insurance/payer, length of stay, diagnosis count, substance-use flags). A templating step (Phase 1) renders each admission as a short natural-language description paired with an instruction and a single-token answer—Yes or No for in-hospital mortality—using eight distinct instruction templates. The resulting corpus contains 50,866 instruction–answer pairs over 49,377 distinct descriptions, with a class balance of 11.4% positive (deaths).

Data integrity

Two problems, invisible from the model’s loss curve, were found by inspecting the Phase 1 output directly.

Validation leakage (critical). All 200 patient descriptions in the held-out validation sample also appeared, verbatim, in the training corpus. As originally written, the pipeline would

have trained on those patients and then reported “validation accuracy” on the same patients, comparing it against a zero-shot baseline that had never seen them—measuring memorization, not generalization, and invalidating the headline comparison. We exclude those patients from training (213 pairs dropped) and verify that 0 of 200 remain in either the training or evaluation split.

Conflicting labels (a data ceiling). 265 descriptions carry *both* Yes and No labels: because the 17 features are coarse, distinct admissions with different outcomes render identical text. No amount of training can separate contradictory examples, so this sets an irreducible noise floor on achievable accuracy.

Honest splitting. Because 1,300 descriptions appear in multiple pairs, we split by description (a grouped split), not at random, yielding 48,653 training and 2,000 evaluation pairs with zero description overlap. Crucially, we rebalance *only* the training split (majority-to-minority ratio 2.0); the evaluation split retains natural prevalence (12% positive), so that evaluation precision is not inflated and agrees with the final 200-patient validation.

Model and quantization

The base model is llama-3.1-8B (8.03B parameters). In fp16 it requires ≈ 16.1 GB, exceeding the 15.9 GB of the target GPU before optimizer state or activations. We therefore load it in 4-bit NF4 with double quantization and fp16 compute (the Turing-class GPU has no bf16), reducing the footprint to 5.6 GB. A LoRA adapter (rank 16, $\alpha=32$, dropout 0.05) is attached to the query and value projections, giving 6,815,744 trainable parameters—0.08% of the model. The base weights stay frozen; only the adapter is trained.

Training

We train for one epoch with an effective batch size of 8 (micro-batch 4, gradient accumulation 2), learning rate 2×10^{-4} with 100 warmup steps, and gradient checkpointing. Sequences are capped at 128 tokens with dynamic per-batch padding (measured median length 97), and prompt tokens are masked to -100 so the loss is computed on the answer token alone—a change that concentrates the gradient on the prediction rather than on regurgitating boilerplate. We monitor a held-out eval split every 150 steps on accuracy, recall, precision, and F1, and keep the best checkpoint by F1.

Evaluation protocol

Final evaluation uses a 200-patient sample held out entirely from training. We compare the untuned base model at zero-shot against the same base with the LoRA adapter, both decoded greedily at **temperature 0.0**. Determinism is not a tuning preference here but a requirement: at temperature 0.0 the same patient always yields the same assessment, which is what makes the system auditable.

Systems constraints

Two hardware realities shaped the design. *Memory*: with checkpointing off, batch 8×2 exceeded the GPU; the shipped configuration (batch 8×2 , checkpointing on) peaks at 10.73 GB. *Thermal*: the GPU is a passively cooled 70 W datacenter card that, in a desktop chassis, throttles roughly $4 \times$ within about 100 seconds of sustained load. A sequence of improvised interventions—an air duct, a floor fan ($\approx \$30$), a 60 W power cap, and a batch-size reduction—held it at a stable 94 °C, about 2 °C inside the hardware shutdown threshold, for the full run.

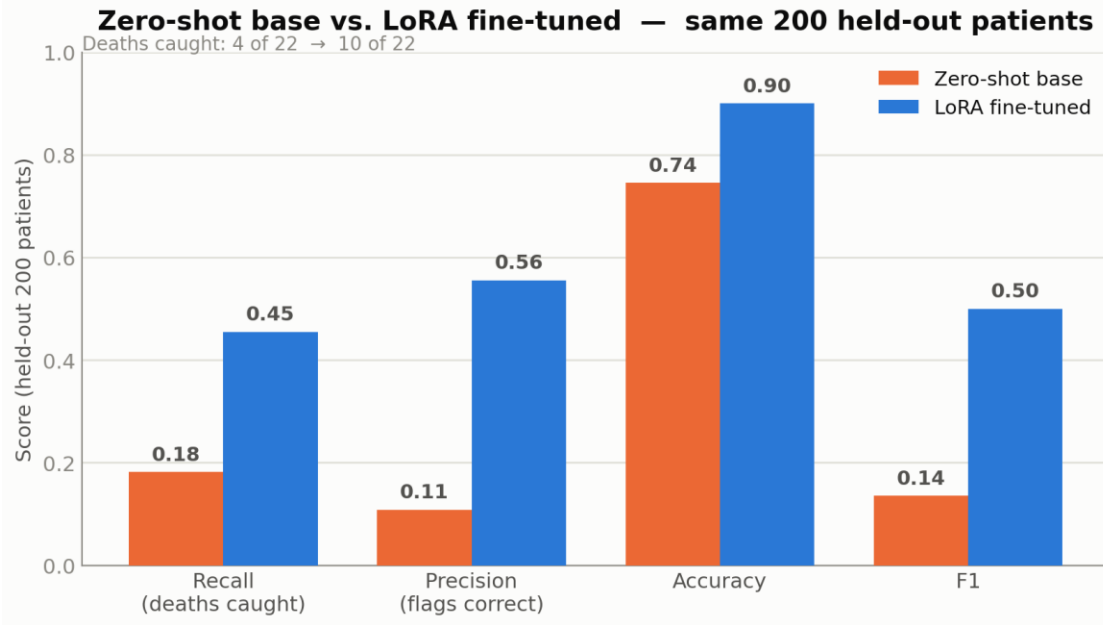
Results

Held-out performance

Table 1 and Figure 2 report the headline result on the 200 held-out patients. The adapted model improves on the zero-shot base on every metric simultaneously: it catches 2.5× as many deaths and is roughly 5× more precise when it does flag a patient. It did not trade one axis for the other, and it did not collapse into the majority class.

Held-out results on 200 patients never seen in training. Both models are the same Llama-3.1-8B; only the 0.08% adapter differs.

Metric	Zero-shot base	LoRA fine-tuned	Change
Recall (deaths caught)	0.18 (4/22)	0.46 (10/22)	2.5×
Precision (flags correct)	0.11	0.56 (10/18)	~5×
Accuracy	74.5%	90.0%	+15.5 pt
Deaths flagged	4 of 22	10 of 22	+6



Zero-shot base vs. LoRA fine-tuned on the same 200 held-out patients, both at temperature 0.0. Improvement is on both axes at once.

Confusion matrix

Figure 3 shows the held-out confusion matrix for the adapted model. Of 22 true deaths it catches 10 (true positives) and misses 12; of 178 survivors it false-alarms on only 8. The total number of patients flagged (18) sits near the true death count (22), indicating that the model learned the correct *scale* of the problem—a calibrated, usable operating point rather than a model that flags everyone or no one.

Held-out confusion matrix — LoRA fine-tuned

Actual survive	170 96% of row True negative	8 4% of row False alarm
Actual die	12 55% of row Missed death	10 45% of row True positive
	Predicted survive	Predicted die

200 patients · 22 deaths, 10 caught · 18 flagged, 8 false alarms · recall 0.45 · precision 0.56 · accuracy 90%

Held-out confusion matrix (LoRA). Color encodes each cell's share of its row so both rows remain legible despite the 170-vs-rest class imbalance.

Training dynamics as an operating frontier

Across one epoch the run passed through four visible phases (from the run's evaluation log): an initial *collapse* (recall 0, hiding in the base rate); a *breakout* in which class-balanced training pulled the model off the base rate and recall oscillated violently; a *damping* phase as the learning rate decayed; and *convergence*, with F1 peaking and the flag count settling near the true death count. The oscillation is itself a result. At two adjacent steps the same weights presented opposite operating points—an aggressive point (recall 0.51, precision 0.26, many flags) and a conservative one (recall 0.15, precision 0.69, few flags). The model did not get better or worse between them; it moved along its own recall–precision frontier.

Efficiency

The entire result was obtained by training 6.8M parameters (0.08% of the model), on a 5.6 GB quantized base, on a single GPU, for an added hardware cost of about \$30, with fully deterministic inference. The medical knowledge that made it work was already frozen in the pretrained weights; fine-tuning *unlocked* it rather than adding it.

Discussion

The operating point is a values decision.

Because the same weights can be read at many thresholds, choosing where to operate—how many false alarms are acceptable to catch one more death—is not a technical default

to be chosen silently by a library. It is a clinical-values decision that belongs to an accountable institution. Reporting a single number hides exactly the choice that governance should own.

The ceiling is the data, not the model.

The residual oscillation never fully vanished because the limit is in the features: contradictory labels (Section 3) mean no decision boundary can be stable. This points directly at the highest-value improvement: richer features (raw labs, vital signs, diagnosis codes) that separate patients the current descriptions collapse together. The bottleneck is feature resolution, not model capacity or method.

A fairness note on proxies.

Feature selection dropped race but retained insurance/payer. In the MIMIC cohort, payer encodes race and class; a model that omits race but keeps its proxy can appear fair on a checklist while distributing its errors along the axis the omission was meant to protect. Whether to remove the proxy, keep and flag it, or fund the intervention the disparity implies is a policy question, not a technical one.

Energy was spent upstream.

The large energy cost of this capability was front-loaded into pretraining, absorbed by whoever trained the base model and inherited here for free. Fine-tuning to beat that base cost near-zero marginal energy. That asymmetry—ecological cost socialized upstream, benefit accruing to every downstream fine-tuner—is itself a governance question about who authorized the expenditure on behalf of everyone who now depends on it.

Limitations.

The headline is a single 200-patient validation sample under one random seed; confidence intervals at this size are wide. The step-level training dynamics come from a single run's log rather than a saved artifact. MIMIC-III is a single institution's historical data, so external validity is unestablished. We report a discrimination operating point but not a calibration analysis.

Conclusion and Future Work

A small custom adapter on a small, open foundation model beat that model's own zero-shot baseline on in-hospital mortality prediction—improving recall, precision, and accuracy at once—at minimal compute and energy cost and with deterministic, auditable output. Just as important as the numbers is the discipline behind them: a validation leak that would have invalidated the claim was found and removed, the performance ceiling was traced to the data rather than the model, and the operating point was framed as the clinical-values decision it is.

Future work follows directly. *Feature enrichment*—replacing coarse counts with raw labs, vitals, and diagnosis codes—is the single highest-value step for lowering the conflicting-label noise floor. We would report the full recall–precision frontier across checkpoints rather than a single point, run an *ensemble check* against structured baselines (gradient-boosted trees, AUC 0.757; a deep network, AUC 0.754) to test whether the text model misses *different* patients, and add calibration and external validation. A planned Phase 3 explores a retrieval-augmented, clinical-persona architecture as a third comparison point. If redone, we would save per-step evaluation artifacts from the outset and budget for adequate GPU cooling before, not during, the run.

9

E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Int. Conf. on Learning Representations (ICLR)*.

T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*.

A. E. W. Johnson, T. J. Pollard, L. Shen, L. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific Data* 3, 160035.

L. Y. Jiang, X. C. Liu, N. P. Nejatian, et al. 2023. Health system-scale language models are all-purpose prediction engines. *Nature* 619, 357–362.

L. Rasmy, Y. Xiang, Z. Xie, C. Tao, and D. Zhi. 2021. Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine* 4, 86.

A. Grattafiori et al. (Llama Team, AI @ Meta). 2024. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.